



Sharif University of Technology
Department of Electrical Engineering

Course Project: Introduction to Machine Learning

Phase 1: Out-of-Distribution Detection with GMM and EM

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Contents

1	Question 1: Single-Component Gaussian Model Analysis	2
1.1	1. Explicit Derivation of the Log-Likelihood Function	2
1.2	2. Geometric Interpretation and Ellipsoid Proof	2
1.3	3. Exact Equation of the Boundary	3
2	Question 2: Gaussian Mixture Models and Jensen's Inequality	4
2.1	1. The Log-Sum Problem	4
2.2	2. Deriving the Lower Bound (Jensen's Inequality)	4
2.3	3. Relation to the EM Objective Function	5
3	Question 3: M-Step Derivation for Means	6
3.1	1. The Optimization Objective	6
3.2	2. Gradient Computation and Optimization	6
3.3	3. Interpretation as Weighted Average	7
4	Question 4: High-Dimensional Behavior and The Curse of Dimensionality	8
4.1	1. Expected Log-Likelihood of ID Data	8
4.2	2. The Paradox of High-Likelihood OOD Samples	8
4.3	3. Why Likelihood is Unreliable in High Dimensions	9

1 Question 1: Single-Component Gaussian Model Analysis

1.1 1. Explicit Derivation of the Log-Likelihood Function

We consider the in-distribution (ID) data modeled by a d -dimensional Multivariate Gaussian distribution $p_{ID}(x) = \mathcal{N}(x; \mu, \Sigma)$. The probability density function is given by:

$$p_{ID}(x) = \frac{1}{(2\pi)^{d/2}(\det \Sigma)^{1/2}} \exp \left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu) \right) \quad (1)$$

To derive the log-likelihood function $\mathcal{L}(x) = \ln p_{ID}(x)$, we apply the natural logarithm to both sides. Using the property $\ln(abc) = \ln a + \ln b + \ln c$:

$$\begin{aligned} \ln p_{ID}(x) &= \ln((2\pi)^{-d/2}) + \ln((\det \Sigma)^{-1/2}) + \ln \left(\exp \left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu) \right) \right) \\ &= -\frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln(\det \Sigma) - \frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu) \end{aligned}$$

Thus, the explicit log-likelihood function is:

$$\boxed{\ln p_{ID}(x) = -\frac{1}{2} (d \ln(2\pi) + \ln |\Sigma| + (x - \mu)^\top \Sigma^{-1}(x - \mu))} \quad (2)$$

1.2 2. Geometric Interpretation and Ellipsoid Proof

The detection rule classifies a sample x as ID if its log-likelihood exceeds a threshold τ :

$$\ln p_{ID}(x) \geq \tau \quad (3)$$

Substituting Eq. (2) into the inequality:

$$-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu) - \frac{1}{2} (d \ln(2\pi) + \ln |\Sigma|) \geq \tau \quad (4)$$

We isolate the quadratic form $(x - \mu)^\top \Sigma^{-1}(x - \mu)$, which corresponds to the squared **Mahalanobis distance** ($D_M^2(x)$). Multiplying the inequality by -2 reverses the sign ($\geq$ becomes $\leq$):

$$(x - \mu)^\top \Sigma^{-1}(x - \mu) \leq -2\tau - d \ln(2\pi) - \ln |\Sigma| \quad (5)$$

Why is this an Ellipsoid?

Since Σ is a symmetric positive definite (SPD) matrix, it admits an eigendecomposition $\Sigma = U\Lambda U^\top$, where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ contains positive eigenvalues. Consequently, $\Sigma^{-1} = U\Lambda^{-1}U^\top$. Letting $y = U^\top(x - \mu)$ be the coordinates in the principal component basis, the inequality becomes:

$$y^\top \Lambda^{-1} y = \sum_{i=1}^d \frac{y_i^2}{\lambda_i} \leq C \quad (6)$$

This is the canonical equation of a hyper-ellipsoid. Therefore, the condition $\ln p_{ID}(x) \geq \tau$ is geometrically equivalent to requiring x to lie **inside or on the boundary** of this ellipsoid centered at μ .

1.3 3. Exact Equation of the Boundary

The boundary of the classification region is defined where the inequality holds as an equality. From Eq. (5), the exact equation of the ellipsoid surface in terms of μ , Σ , and τ is:

$$(x - \mu)^\top \Sigma^{-1} (x - \mu) = \mathcal{R}^2 \quad (7)$$

where the squared radius (or level set constant) is:

$$\boxed{\mathcal{R}^2 = -2\tau - d \ln(2\pi) - \ln(\det \Sigma)} \quad (8)$$

2 Question 2: Gaussian Mixture Models and Jensen's Inequality

2.1 1. The Log-Sum Problem

For a Gaussian Mixture Model (GMM) with K components, parameters $\Theta = \{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K$, and a given sample x , the log-likelihood is:

$$\ell(\Theta) = \ln \left(\sum_{k=1}^K \pi_k \mathcal{N}(x; \mu_k, \Sigma_k) \right) \quad (9)$$

We are asked to show why this cannot be written as a sum of individual log-likelihoods.

- **Mathematical Observation:** The logarithm function is non-linear. Specifically, for a sum of terms $a_k > 0$, we have the strict inequality:

$$\ln \left(\sum_{k=1}^K a_k \right) \neq \sum_{k=1}^K \ln(a_k) \quad (10)$$

- **Implication:** In Eq. (9), the summation operation is *inside* the logarithm. This prevents the logarithm from acting directly on the exponential function within the Gaussian density $\mathcal{N}(x; \mu_k, \Sigma_k)$.
- **Consequence:** Because of this structure, the parameters of different components (μ_k, Σ_k) are coupled in the summation. Taking the derivative with respect to any parameter results in a complex non-linear equation involving all other components, making a closed-form analytical solution (like in the single Gaussian case) impossible.

2.2 2. Deriving the Lower Bound (Jensen's Inequality)

To overcome the intractability of direct maximization, we derive a lower bound. Let us introduce a latent variable $z \in \{1, \dots, K\}$ and an arbitrary probability distribution $q(z)$ over these latent variables, such that $q_k = q(z = k) \geq 0$ and $\sum_k q_k = 1$.

We can rewrite the log-likelihood by multiplying and dividing each term by q_k :

$$\ln p(x) = \ln \left(\sum_{k=1}^K q_k \frac{\pi_k \mathcal{N}(x; \mu_k, \Sigma_k)}{q_k} \right) \quad (11)$$

This expression can be viewed as the logarithm of an expectation with respect to the distribution q :

$$\ln p(x) = \ln \left(\mathbb{E}_q \left[\frac{\pi_k \mathcal{N}(x; \mu_k, \Sigma_k)}{q_k} \right] \right) \quad (12)$$

Jensen's Inequality states that for a concave function f (such as $\ln(\cdot)$) and a random variable X :

$$f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)] \quad (13)$$

Applying this to our equation:

$$\ln \left(\sum_{k=1}^K q_k \frac{\pi_k \mathcal{N}(x; \mu_k, \Sigma_k)}{q_k} \right) \geq \sum_{k=1}^K q_k \ln \left(\frac{\pi_k \mathcal{N}(x; \mu_k, \Sigma_k)}{q_k} \right) \quad (14)$$

The Right Hand Side (RHS) is the derived lower bound, often denoted as the Evidence Lower Bound (ELBO):

$$\mathcal{L}(q, \Theta) = \sum_{k=1}^K q_k (\ln \pi_k + \ln \mathcal{N}(x; \mu_k, \Sigma_k) - \ln q_k) \quad (15)$$

2.3 3. Relation to the EM Objective Function

This lower bound is the fundamental objective function optimized by the Expectation-Maximization (EM) algorithm.

1. **E-Step (Tightening the Bound):** In the E-step, we maximize $\mathcal{L}(q, \Theta)$ with respect to the distribution q , holding parameters Θ fixed. The optimal q is the posterior probability (responsibility):

$$q_k^* = p(z = k|x; \Theta^{(t)}) = \frac{\pi_k \mathcal{N}(x; \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x; \mu_j, \Sigma_j)} \quad (16)$$

At this point, the lower bound touches the true log-likelihood function (the inequality becomes an equality).

2. **M-Step (Maximizing the Bound):** In the M-step, we maximize $\mathcal{L}(q^*, \Theta)$ with respect to Θ . Notice that in Eq. (14), the logarithm is now *inside* the summation. This means $\ln$ acts directly on $\mathcal{N}$, simplifying the exponential and decoupling the parameters.

$$Q(\Theta, \Theta^{(t)}) = \mathbb{E}_{z|x, \Theta^{(t)}} [\ln p(x, z|\Theta)] = \sum_{k=1}^K \gamma_k \ln (\pi_k \mathcal{N}(x; \mu_k, \Sigma_k)) \quad (17)$$

This function Q is exactly the expected complete-data log-likelihood, which allows for closed-form updates (weighted means and covariances).

Therefore, the EM algorithm works by iteratively constructing this local lower bound (E-step) and then maximizing it (M-step) to monotonically increase the likelihood.

3 Question 3: M-Step Derivation for Means

3.1 1. The Optimization Objective

In the M-step of the EM algorithm, we aim to find the parameters Θ that maximize the expected complete-data log-likelihood, denoted as the Q -function:

$$Q(\Theta, \Theta^{(t)}) = \sum_{i=1}^n \sum_{k=1}^K \gamma_{ik}^{(t)} \ln(\pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k)) \quad (18)$$

To derive the update rule for the mean of the k -th component (μ_k), we isolate the terms in the summation that depend on μ_k . All other terms (involving π_k , Σ_k , or indices $j \neq k$) are treated as constants with respect to optimization over μ_k . Let $J(\mu_k)$ be the objective function strictly for μ_k :

$$J(\mu_k) = \sum_{i=1}^n \gamma_{ik}^{(t)} \ln \mathcal{N}(x_i; \mu_k, \Sigma_k) \quad (19)$$

Expanding the Gaussian log-density (ignoring constants independent of μ_k):

$$\ln \mathcal{N}(x_i; \mu_k, \Sigma_k) = -\frac{1}{2}(x_i - \mu_k)^\top \Sigma_k^{-1}(x_i - \mu_k) + \text{const} \quad (20)$$

Substituting this back into $J(\mu_k)$:

$$J(\mu_k) = -\frac{1}{2} \sum_{i=1}^n \gamma_{ik}^{(t)} (x_i - \mu_k)^\top \Sigma_k^{-1}(x_i - \mu_k) \quad (21)$$

3.2 2. Gradient Computation and Optimization

To find the optimal μ_k , we compute the gradient of Eq. (21) with respect to the vector μ_k and set it to zero. We use the matrix calculus identity for quadratic forms: $\nabla_v[(x - v)^\top A(x - v)] = -2A(x - v)$ (assuming symmetric A). Here, $v = \mu_k$ and $A = \Sigma_k^{-1}$.

$$\begin{aligned} \nabla_{\mu_k} J &= -\frac{1}{2} \sum_{i=1}^n \gamma_{ik}^{(t)} \nabla_{\mu_k} [(x_i - \mu_k)^\top \Sigma_k^{-1}(x_i - \mu_k)] \\ &= -\frac{1}{2} \sum_{i=1}^n \gamma_{ik}^{(t)} [-2\Sigma_k^{-1}(x_i - \mu_k)] \\ &= \sum_{i=1}^n \gamma_{ik}^{(t)} \Sigma_k^{-1}(x_i - \mu_k) \end{aligned}$$

Setting the gradient to zero for the optimal solution $\mu_k^{(t+1)}$:

$$\Sigma_k^{-1} \sum_{i=1}^n \gamma_{ik}^{(t)} (x_i - \mu_k^{(t+1)}) = 0 \quad (22)$$

Since the covariance matrix Σ_k (and thus its inverse) is positive definite, we can pre-multiply by Σ_k to eliminate the matrix term:

$$\sum_{i=1}^n \gamma_{ik}^{(t)} (x_i - \mu_k^{(t+1)}) = 0 \quad (23)$$

Distributing the summation:

$$\sum_{i=1}^n \gamma_{ik}^{(t)} x_i - \mu_k^{(t+1)} \sum_{i=1}^n \gamma_{ik}^{(t)} = 0 \quad (24)$$

Defining the effective number of points in cluster k as $N_k^{(t)} = \sum_{i=1}^n \gamma_{ik}^{(t)}$, we solve for $\mu_k^{(t+1)}$:

$$\boxed{\mu_k^{(t+1)} = \frac{1}{N_k^{(t)}} \sum_{i=1}^n \gamma_{ik}^{(t)} x_i} \quad (25)$$

3.3 3. Interpretation as Weighted Average

The derived update rule has a clear intuitive interpretation:

- It represents a **weighted average** of all data points in the dataset.
- The weight for each data point x_i is its responsibility $\gamma_{ik}^{(t)}$, which represents the probability that x_i belongs to component k .
- Points that are very likely to belong to cluster k (high γ_{ik}) have a strong influence on the new mean, while points far away (low γ_{ik}) have negligible influence.
- This generalizes the standard arithmetic mean used in K-Means (where weights are binary 0 or 1) to a probabilistic framework (soft assignment).

4 Question 4: High-Dimensional Behavior and The Curse of Dimensionality

4.1 1. Expected Log-Likelihood of ID Data

We assume the ID data follows a d -dimensional Gaussian distribution with isotropic covariance $\Sigma = \sigma^2 I$:

$$p(x) = \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{1}{2\sigma^2}\|x\|^2\right) \quad (26)$$

The log-likelihood function is:

$$\ln p(x) = -\frac{d}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|x\|^2 \quad (27)$$

We aim to find the expectation $\mathbb{E}_{x \sim p}[\ln p(x)]$. Using the linearity of expectation:

$$\mathbb{E}[\ln p(x)] = -\frac{d}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \mathbb{E}[\|x\|^2] \quad (28)$$

Since $x \sim \mathcal{N}(0, \sigma^2 I)$, the squared norm $\|x\|^2 = \sum_{i=1}^d x_i^2$ follows a scaled Chi-squared distribution. The expectation of each component is the variance $\mathbb{E}[x_i^2] = \sigma^2$. Thus:

$$\mathbb{E}[\|x\|^2] = \sum_{i=1}^d \mathbb{E}[x_i^2] = d\sigma^2 \quad (29)$$

Substituting this back yields the expected log-likelihood:

$$\mathbb{E}[\ln p(x)] = -\frac{d}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}(d\sigma^2)$$

$$\mathbb{E}[\ln p(x)] = -\frac{d}{2} \ln(2\pi\sigma^2) - \frac{d}{2}$$

4.2 2. The Paradox of High-Likelihood OOD Samples

The question asks how an OOD sample can have a higher likelihood than ID samples ($\ln p(x_{OOD}) > \ln p(x_{ID})$).

From the Gaussian PDF, the log-likelihood is strictly decreasing with respect to the norm $\|x\|$. The maximum likelihood is achieved at the mean ($x = 0$).

- **ID Samples Location (Typical Set):** As derived in Part 1, ID samples do not cluster around the mean (0). Instead, they concentrate in a thin shell (annulus) with radius $r \approx \sqrt{d}\sigma$. This is because in high dimensions, the volume of space scales as r^d , making the volume near the center negligible compared to the volume of the shell.
- **OOD Samples Location:** "Regular structure" (e.g., constant color images, simple patterns) often implies low information content and low variance. In the feature space, these samples often map very close to the origin (mean).

Therefore, if an OOD sample x_{OOD} has a norm smaller than the expected ID norm ($\|x_{OOD}\| < \sqrt{d}\sigma$), it will sit in a higher-density region of the Gaussian distribution, resulting in a higher likelihood score, despite being semantically different.

4.3 3. Why Likelihood is Unreliable in High Dimensions

This analysis highlights a fundamental limitation of density-based methods in high-dimensional spaces:

1. **Density vs. Probability Mass:** Likelihood measures *density*, not probability mass. A point (like the mean) can have extremely high density but near-zero probability mass because the surrounding volume is infinitesimal.
2. **The Typical Set:** A reliable OOD detector should check if a sample belongs to the *Typical Set* (the region covering most of the probability mass), not just if it has high density.
3. **Conclusion:** In high dimensions, the "safe" ID region is a shell. A simple likelihood threshold $\ln p(x) \geq \tau$ corresponds to a ball $\|x\| \leq R$. This accepts everything inside the ball, including the OOD samples near the center that do not resemble the noisy/complex structure of valid ID data.